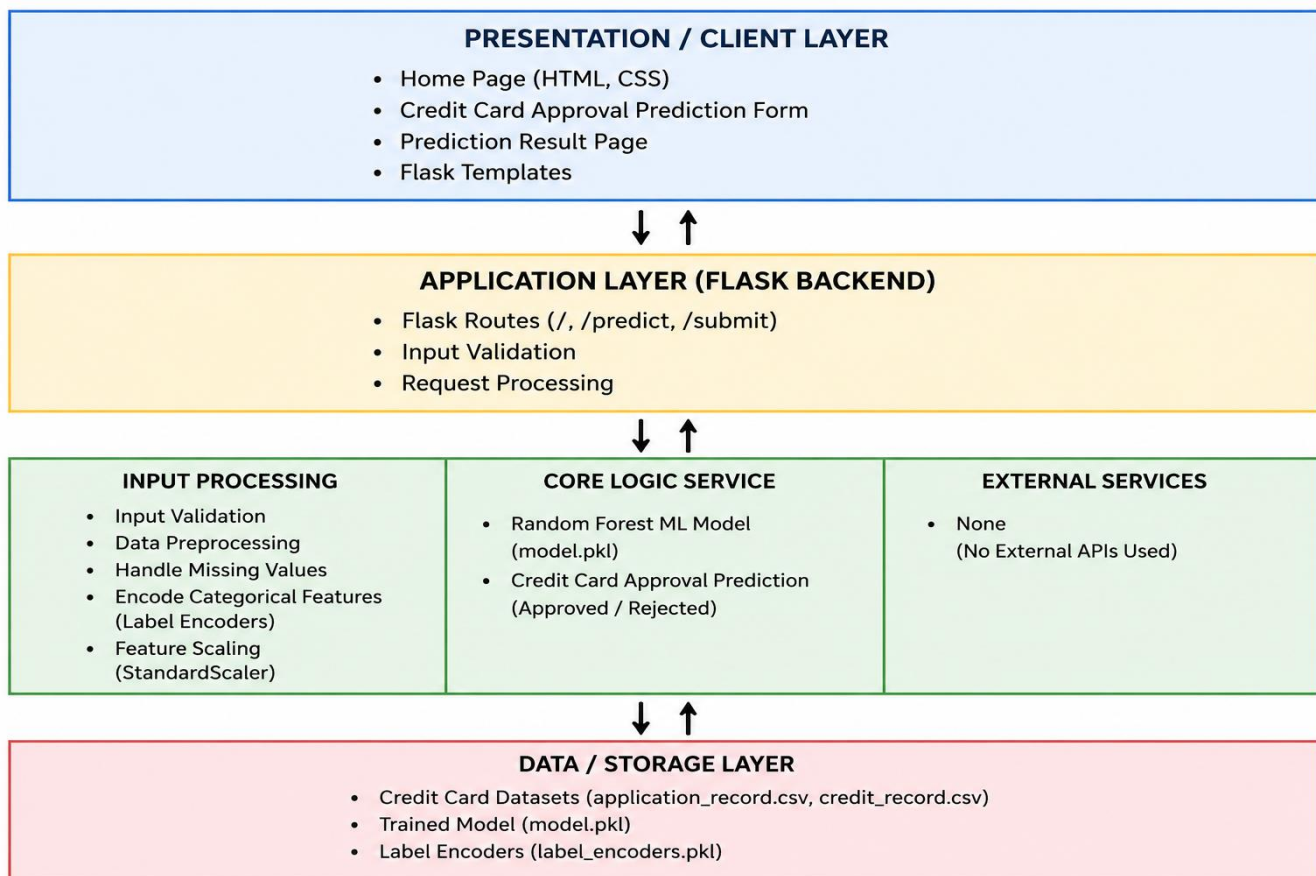


Solution Architecture

Date	27 June 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

The Credit Card Approval Prediction System follows a three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer. The Flask application processes user input and interacts with the trained Machine Learning model to generate quick and accurate credit card approval predictions.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects applicant details and displays the credit card approval prediction through web pages.	HTML5, CSS3, Flask Templates
Application Layer (Flask Backend)	Handles routing, input validation, request processing, and communicates with the Machine Learning model.	Python, Flask
Core Logic Service	Encodes applicant data and predicts credit card approval using the trained Random Forest model.	Scikit-learn, Joblib
Data / Storage Layer	Stores the credit card datasets, trained model, and label encoders required for prediction.	CSV Dataset, model.pkl, label_encoders.pkl